where

$$\mathrm{d}Z_i = \sum_{j=1}^{2\mathrm{N}_c} \frac{\partial Z_i}{\partial z_j} \mathrm{d}z_j.$$

Taking the derivative with respect to t and defining $\dot{Z}_i = f_i(\boldsymbol{Z}(t, \boldsymbol{z}))$, we can see that

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\mathrm{d}Z_i &= \sum_{j=1}^{2\mathrm{N}_c} \frac{\partial \dot{Z}_i}{\partial z_j} \mathrm{d}z_j \\ &= \mathrm{d}\dot{Z}_i. \end{aligned}$$

The symplecticness condition therefore becomes

$$\mathrm{d}\boldsymbol{Z} \wedge \boldsymbol{J}\mathrm{d}\dot{\boldsymbol{Z}} + \mathrm{d}\dot{\boldsymbol{Z}} \wedge \boldsymbol{J}\mathrm{d}\boldsymbol{Z} = 0,$$

or, in canonical coordinates,

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}\dot{q}_j \wedge \mathrm{d}p_j + \mathrm{d}q_j \wedge \mathrm{d}\dot{p}_j = 0,$$

which is essentially a Leibniz differentiation rule for wedge products. The functions $\dot{q}_j, \dot{p}_j$ that appear here are just the components of the vector field.

We next demonstrate that a similar property holds for the constrained equations. Taking differentials, we have

$$\begin{aligned} \mathrm{d}\dot{\boldsymbol{q}} &= \mathrm{d}\boldsymbol{M}^{-1}\boldsymbol{p}, \\ \mathrm{d}\dot{\boldsymbol{p}} &= -\mathrm{d}\nabla U(\boldsymbol{q}) - \sum_{k=1}^{l} \mathrm{d}(\lambda_k \nabla g_k(\boldsymbol{q})), \\ 0 &= \mathrm{d}g_j(\boldsymbol{q}), \qquad j = 1, 2, \ldots, l, \end{aligned}$$

Next, we interchange time derivatives and differentials, and use the chain rule to write

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{d}\boldsymbol{q} = \boldsymbol{M}^{-1}\mathrm{d}\boldsymbol{p}, \tag{4.11}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{d}\boldsymbol{p} = -U''(\boldsymbol{q})\mathrm{d}\boldsymbol{q} - \sum_{k=1}^{l} [(\mathrm{d}\lambda_k)\nabla g_k(\boldsymbol{q}) + \lambda_k g_k'' \mathrm{d}\boldsymbol{q}], \tag{4.12}$$

$$0 = g_j'(\boldsymbol{q})\mathrm{d}\boldsymbol{q}, \qquad j = 1, 2, \ldots, l, \tag{4.13}$$